

HPAI modelling challenge model description — second phase

Team: Flockbusters

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Model name: *County-level patch SEIR model of HPAI transmission in Jolly Island*

Version: 2

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Repository / code link (optional): <https://github.com/hamfat/HPAI-ChallengeModel> description

You may indicate only the changes since the previous phase, describe the modifications made, or, if nothing has changed, state this here.

Model overview

Please provide the following key information where relevant for the model used:

- **Type:** Mechanistic, compartmental, network-based metapopulation model. We have implemented environmental reservoir and production-type structure to it.
- **Stochastic vs deterministic:** stochastic
- **Timestep:** Discrete time, with a daily time step. The model is simulated forward day by day.
- **Entities and levels considered:** (i) farm level: farms are the epidemiological units represented within compartments. (ii) Production type level: we have three structured subpopulations per county: Broiler 2 (B2), Duck (D), and Other (O). (iii) Finally country Level: each county is a node (patch) in the network. Also each county has a shared environmental viral-load compartment.
- **Spatial scale:** Metapopulation (multi-patch network), where a node is coupled via a directed weighted transmission network.

- **Interaction mechanisms:** We have transmission occur through multiple pathways.
 - (i) Within-county (local) transmission
 - (ii) Network mediated between-county transmission, governed by weighted adjacency matrix W_{ij} .
 - (iii) County-level environmental contamination and exposure.
- **Implementation:** we have used language with “tidyverse”, “sf”, “ggplot2”, “lubridate”, “patchwork” packages. We have not used any epidemiological modelling packages and have implemented the code from scratch. We have also implemented some simulations in Python using array-based stochastic simulation. We have used standard packages like numPy, pandas, matplotlib, and geopandas, with repeated ensemble runs to quantify uncertainty.

Please provide a short narrative describing the model structure and key assumptions (please provide a diagram, if relevant):

We have developed a spatially explicit, stochastic SEIR-type metapopulation model to describe the spread of the HPAI virus across counties. Each county is treated as a node (patch) in a network and contains three production-type subpopulations: Broiler 2 (B2), Duck (D), and Other (O). Within each county and production type, farms are grouped into epidemiological compartments representing disease status.

We have let transmission occur through three interacting mechanisms: (i) within-county local transmission among farms, (ii) environmental transmission via a shared county-level viral reservoir, and (iii) between-county transmission realised by time-varying movement data. The environmental compartment accumulates viral shedding from infectious farms and decays over time. These indicate indirect transmission pathways.

We have simulated the model in discrete daily time steps using stochastic transitions. We have generated new infections via probabilistic processes, and disease progression and removal through binomial transitions. Also, we have a time-dependent confinement which reduces environmental exposure, shedding, and movement intensity from a specified date.

Various assumptions that we have included are: (i) farms are aggregated rather than individual entities, (ii) populations within each county are mixed, (iii) movement data adequately captures between-county transmission pathways, (iv) the environment is homogeneous within each county.

Describe state variables:

name	symbol	description
Susceptible farms	$S_{k,i}$	Farms of production type k in county i that are healthy and can become infected
Exposed farms	$E_{k,i}$	Farms that are infected but not yet infectious
Infectious farms	$I_{k,i}$	Farms actively shedding virus and contributing to transmission.
Removed farms	$R_{k,i}$	Farms that are no longer infectious due to recovery, culling, or death
Environmental viral load	B_i	Accumulated virus in the environment within county i , contributing to indirect transmission

Note that

1. $k \in \{B_2, D, O\}$ denotes production type,
2. $i = 1, \dots, N$ denotes counties,
3. The environmental compartment B_i is shared across production types within a county,
4. This is therefore a structured SEIR model with an environmental reservoir (SEIRB).

Processes and parameter values

For each of the processes below, if relevant to the approach used, describe formulation (text and equations), triggers, randomness, and links to state variables (the list provided is not exhaustive; please feel free to add any additional details).

Demographic processes

- **Entry/exit processes:** Entry is not explicitly modelled. As for exit farms transition to the removed compartment via a stochastic binomial process with daily probability γ representing culling due to infection.
- **Production cycles / occupancy / downtime:** Production types are static categories. We haven't included any production cycles/occupancy/downtime.

- **Movements or exchanges (if any):** We have explicitly modelled movements using time-varying county-to-county movement data. Let $W_{ij}^{(k)}(t)$ denote the number of movements from county j to county i for production type k at time t . The movement-driven force of infection for county i and production type k is given by

$$\lambda_{k,i}^{\text{move}}(t) = \beta_{\text{move}}^{(k)} \sum_{j \neq i} W_{ij}^{(k)}(t) \frac{I_{k,j}(t)}{N_{k,j}}$$

where: $\beta_{\text{move}}^{(k)}$ is the movement transmission coefficient, and $I_{k,j}(t)/N_{k,j}$ is the infectious prevalence in source county j .

Epidemiological processes

- **Health states:** same as in the model overview

Transmission: Transmission occurs through three interactive pathways

(I) Within-county local transmission: For county i and production type k the local force of infection is

$$\lambda_{k,i}^{\text{loc}}(t) = \beta_{\text{loc}}^{(k)} I_{k,i}(t).$$

(II) Between-county transmission: The model uses time-varying movement data. The movement-related force of infection is

$$\lambda_{k,i}^{\text{move}}(t) = \beta_{\text{move}}^{(k)} \sum_{j \neq i} W_{ij}^{(k)}(t) \frac{I_{k,j}(t)}{N_{k,j}}.$$

(III) Environmental transmission: Environmental transmission is mediated by the county-level viral load $B_i(t)$:

$$\lambda_{k,i}^{\text{env}}(t) = \beta_{\text{env}}^{(k)} B_i(t) c_k^{\text{exp}}(t),$$

where $c_k^{\text{exp}}(t)$ is a production-specific exposure scaling factor. This allows reduced environmental exposure under confinement. The total force of infection is therefore

$$\lambda_{k,i}(t) = \lambda_{k,i}^{\text{loc}}(t) + \lambda_{k,i}^{\text{move}}(t) + \lambda_{k,i}^{\text{env}}(t).$$

New exposures are generated stochastically. The number of new infections entering $E_{k,i}$ is drawn from a Poisson distribution based on the force of infection and the number of susceptible farms.

- **Progression and recovery:** We implement a latent progression

$$E_{k,i} \xrightarrow{\sigma} I_{k,i}$$

through a binomial draw with daily progression probability σ :

$$\text{Prog}_{k,i}(t) \sim \text{Binomial}(E_{k,i}(t), \sigma) .$$

We also have infection removal

$$I_{k,i} \xrightarrow{\gamma} R_{k,i}$$

through a binomial draw with daily progression probability γ :

$$\text{Rem}_{k,i}(t) \sim \text{Binomial}(I_{k,i}(t), \gamma) .$$

- **Detection and surveillance (if modelled):** Even though we do not represent detection as a separate epidemiological state, we use observed data for initialization and calibration of the simulated epidemic trajectory.
- **Disease-induced mortality / production impact (if modelled):** We have done this through the removal compartment representing culling of infected farms.

Disease management (control measures)

- **Measures included** *[please list all measures included in your model - if any - and describe briefly how you implemented each measure]*

Management measure	Included?	Implementation description
National standstill	N	
Suspicion management	N	

Reactive culling	Y	We implicitly included this via transition $I_{k,i} \rightarrow R_{k,i}$ via a binomial process with daily probability γ . This represents removal (culling) of infected farms. No explicit trigger or delay is modeled.
Contact tracing	N	
Zoning: Movement bans in the regulated zones	N	
Zoning: Enhanced biosecurity and detectability in regulated zones	Y (implicit)	We represent this through time-dependent confinement scaling factors applied to environmental exposure, shedding, and movement intensity. This reduces transmission for certain production types (B2, D) after the intervention start date.
Preventive culling	Y	
Pre-movement testing in the HRZ	N	

Parameter table (symbol, description, value, sources)

Populate the table below. If parameters vary by context (species, production type, region, period), indicate the indexing explicitly.

Symbol	description	unit	default value	range / distribution	source
β_{loc}	Within-county transmission coefficient	$\frac{1}{\text{time}}$ per infection farm	Production-specific	Positive real values and calibrated	calibrated in model
β_{env}	Environmental transmission coefficient	$\frac{1}{\text{time}}$ per unit environmental load.	Production-specific	Positive real values	Calibrated
β_{move}	Movement-mediated transmission coefficient for a production type	$\frac{1}{\text{time}}$	Production-specific	Positive real values	Calibrated
W_{ij}	Number of movements from county i to j	Movements per day	Data-driven	Empirical values dependent on time	Movement dataset
σ	Progression rate from exposed to infectious class. The value σ^{-1} is mean latent period.	$\frac{1}{\text{time}}$	1/3	Fixed	Assumed

γ	Removal rate from infectious to removed class. The value γ^{-1} is mean infectious period	$\frac{1}{\text{time}}$	1/7	Fixed	Assumed
$\kappa_{\text{env}}^{(k)}$	Environmental shedding rate from infectious farms of type k	environmental load $\cdot \frac{1}{\text{time}}$ per infectious farm	Production-specific	Positive real values	Assumed
δ_{env}	Environmental decay rate	$\frac{1}{\text{time}}$	0.2	$0 \leq \delta_{\text{env}} \leq 1$	Assumed
$c_k^{\text{exp}}(t)$	Time-dependent environmental exposure scaling factor (confinement effect)	dimensionless	1 before intervention	(0,1] after confinement	Intervention assumption
$c_k^{\text{shed}}(t)$	Time-dependent movement scaling factor (confinement effect)	Dimensionless	1 before intervention	(0,1] after confinement	Intervention assumption
λ_{seed}	Initial seeding intensity (Poisson mean for initial infections)	Farms	0.5	Poisson distribution	assumed
N_i	Total number of farms in county i .	Farms	County and type specific	Fixed	Farm dataset

Parameter estimation

We specified the parameters using a combination of epidemiological assumptions, production-type-specific tuning, and empirical data. We directly used daily movement data to construct time-varying transmission inputs, and used the observed case dates to initialize infection seeding and to define a fitting period. We assessed the behaviour of the model by comparing simulated epidemic trajectories with the observed outbreak progression up to 13 February 2026. Then we used the model for forecasting. We also used ensemble simulation to quantify uncertainty in projected outcomes.

Initial conditions

- **Simulation start date:** 22 December 2025
- **Initial infection seeding:** We seeded observed confirmed cases directly on their confirmation dates by moving one farm into the infectious compartment in the corresponding county and production type. If a production group had no seeded infection, we drew the number of initial infectious farms from a Poisson distribution with mean 0.5.

Simulations and outputs

- **Number of runs / repetitions:** 300 stochastic ensemble runs
- **Random seed handling:** We used reproducible seeds. We ran the ensemble with base seed 2026. We also executed a single test run with seed 0 for checking model behaviour.
- **Outputs:** We show total infectious farms over time, production-specific infectious trajectories, county-level infectious burden over time, county-level spatial snapshots of mean infectious farms, removed farms, environmental viral load. We also show summary statistics including ensemble means and 95% intervals.
- **Output format:** We produce figures and save image files. Outputs include time-series plots and multi-panel spatial maps.

Second period

For each of the requests below, please describe the results you obtained. Keep in mind that you do not have to address all these requests. There is no length limit for this part.

- ***A general description of the ongoing epidemic, including a table describing the distribution of the outbreaks per species and production types, a timeline of the incidence and a visual representation of the spatial distribution of the outbreaks. (please also provide the visual representation as a separate file in pdf format).***

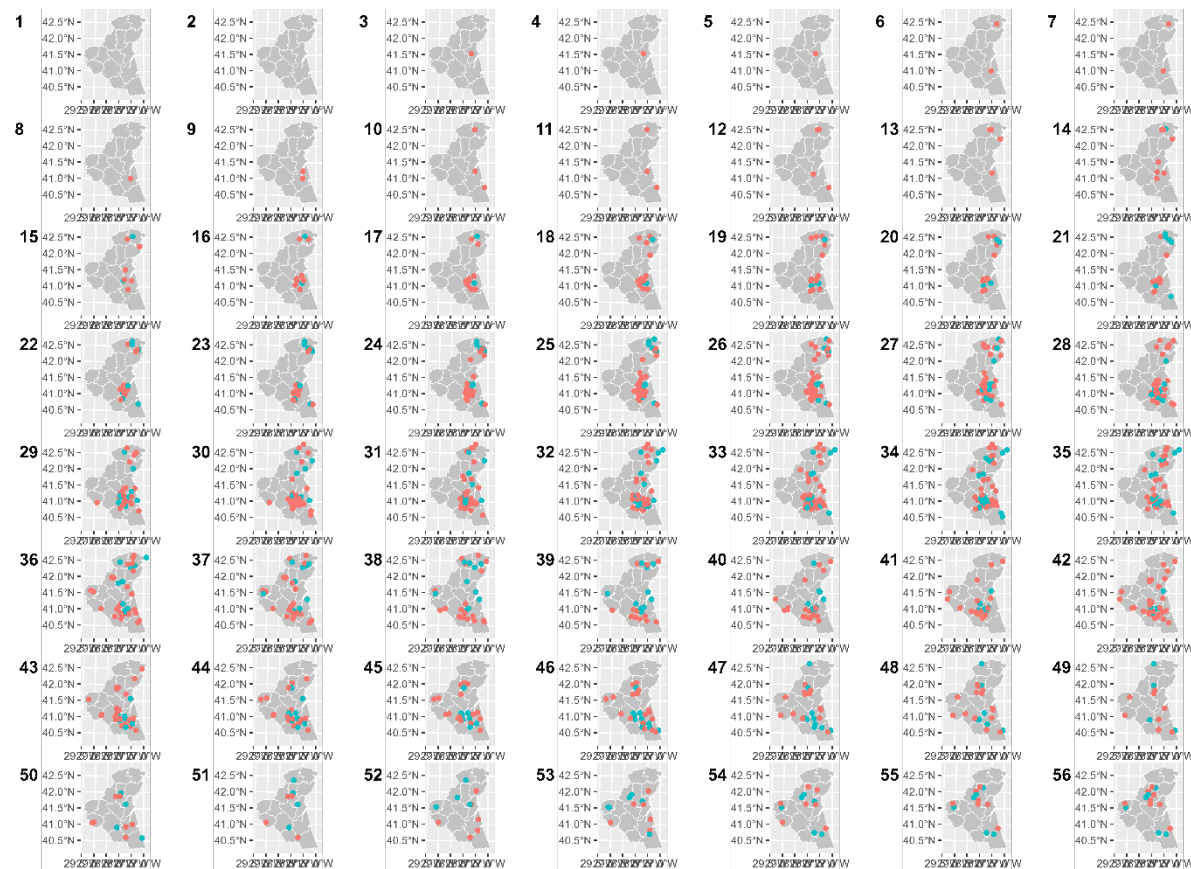
Until the end of the data collection period (13/02/2026), there were a total of 466 confirmed cases. The epidemic primarily affected the chicken farms, which accounted for 77% of the cases. When characterised by production types, it was noted that the outbreaks concentrated in “broiler production systems”, with fewer cases observed in the other types. This suggests that transmission through the broiler farms is mostly associated with intensive production and frequent batch turnover, which increases indirect contacts between farms. We can tell that the epidemic is structured along poultry industry organisations rather than occurring uniformly across all farm types. Geographically speaking, after the end of the first data collection period, the epidemic moved to the western areas, while the situation in the northern and central areas continued to ease.

The table below summarises the distribution of confirmed outbreaks by species and production type. It was the chicken farms that saw the majority of the outbreaks, particularly in the broiler production system. The duck farm outbreaks were relatively lower. This provides an indication that the transmission occurred due to intensive production networks. The production-related contacts might play an important role in the disease spread.

species	production	n_outbreaks
chicken	broiler_1	81
chicken	broiler_2	225
chicken	layer	53
duck	conventional	59
duck	organic	48

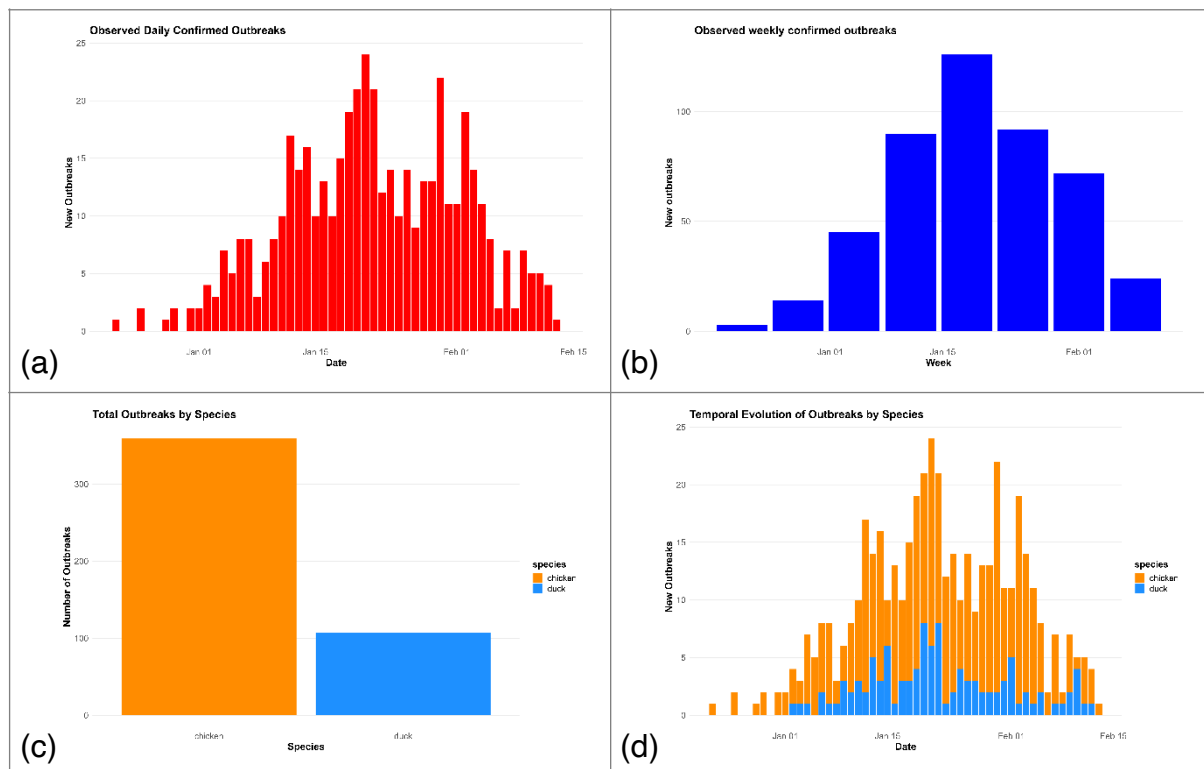
The first figure shows the location of infected farms from the date the first suspicious case was

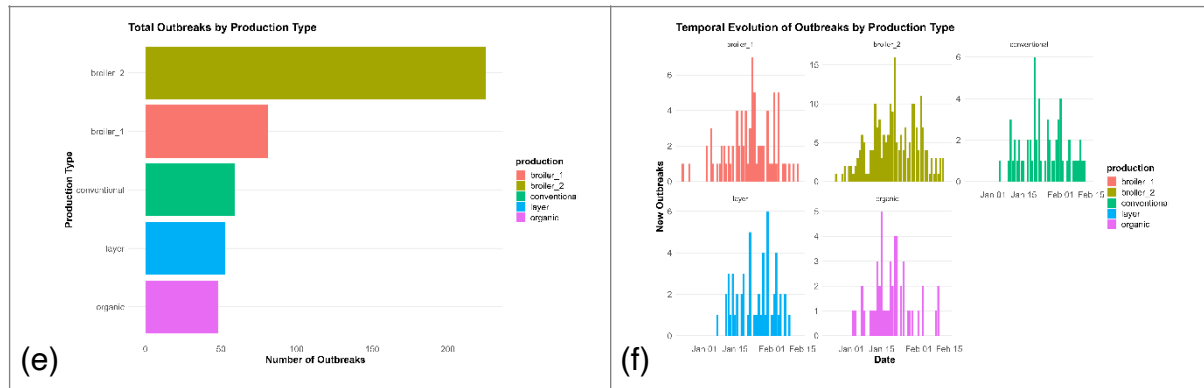
reported (20/12/2025; Day 1) to the end of the data collection period (13/02/2026; Day 56). As seen from the figure, the first confirmed case was found on the East coast. Although reactive culling was completed on Day 5, the virus spread to two other farms, one in the North and one in the South, on Day 6. Since then, the virus eventually spread across the Eastern half of the island, which mainly consisted of two clusters, one in the North and one in the South. The first confirmed case along the West coast was reported on Day 29, and more cases appeared after Day 36 in that area. Meanwhile, the situation in the Central and North areas became better, with the number of confirmed cases continued to decline until the end of the data collection period. Most of the infected farms were breeding chickens (orange), rather than ducks (green).



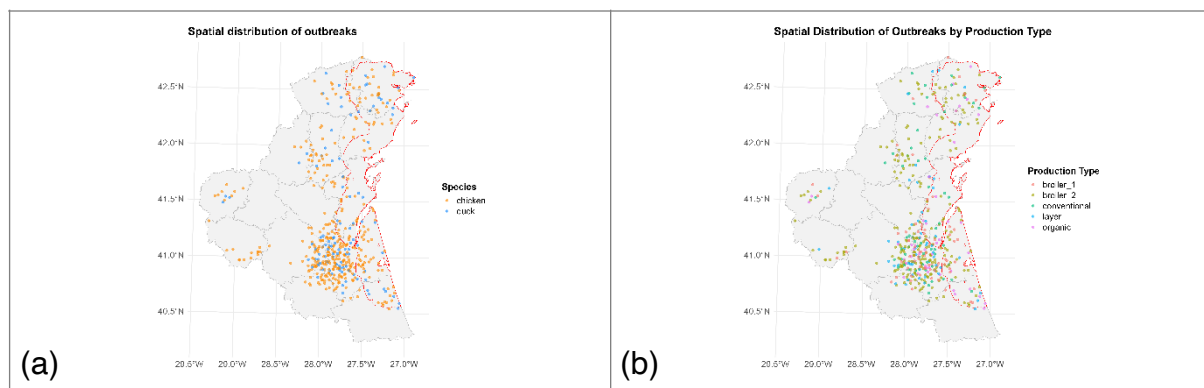
The figures below depict the epidemic by different characteristics. Panel (a) shows a daily incidence curve. We see sporadic cases to start with, followed by rapidly increasing cases, and sustained transmission peaks until mid-January, representing growth in disease spread. The number of confirmed cases then began to drop, although some spikes were observed. We provide a weekly epidemic curve in panel (b). This clearly portrays the weekly aggregated incidence, smoothing reporting variability and revealing a clear structure with a rapid increase in outbreaks over successive weeks until mid-

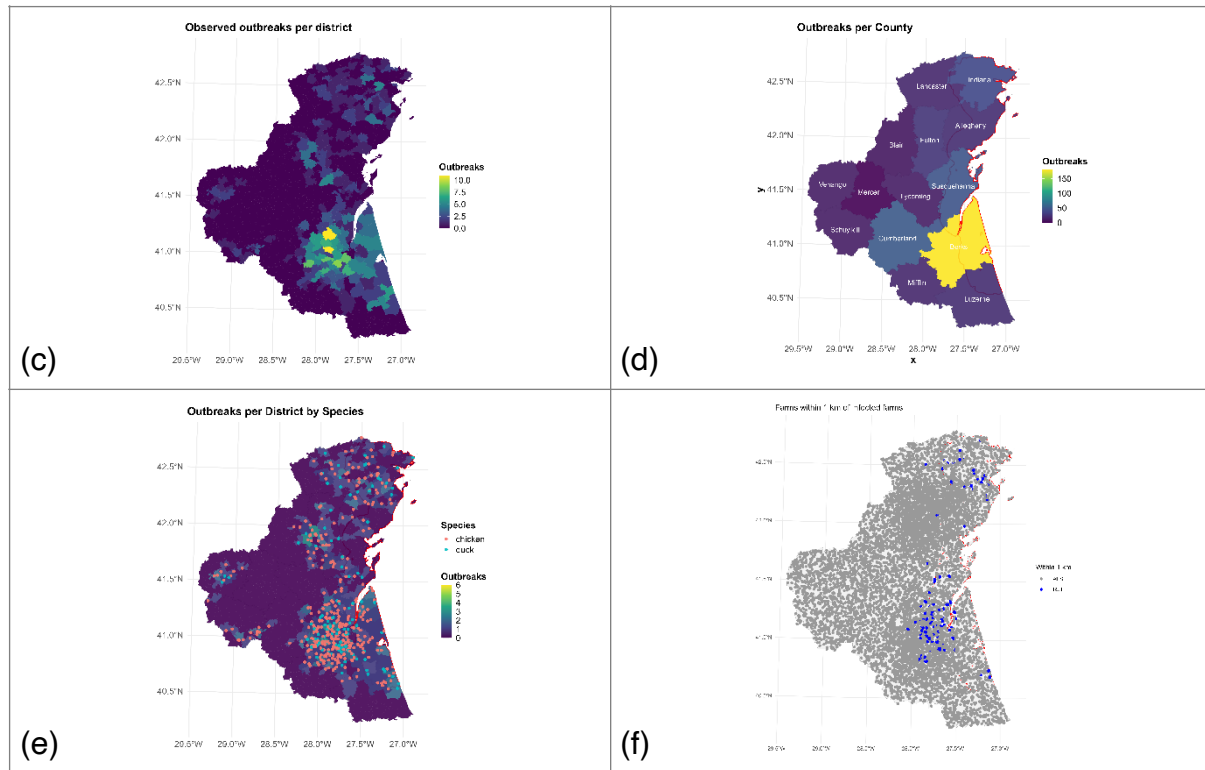
January, and dropped eventually. In panel (c) we show the counts of infected farms by species. We note that the chickens drove the infection transmission, whereas the ducks were affected but much lesser in number. This means, the transmission likely occurs within commercial chicken production networks. We show the temporal evolution of outbreaks by species in panel (d). Chicken outbreaks occur earlier and in higher numbers and duck outbreaks appear later. These suggest a secondary spillover scenario. Next we classify the farms by the production types and present the outbreaks according to the production type in panel (e). We see that the broiler types account for the majority of infections. Finally in panel (f) we show which sector seeded the virus spread, by presenting a plot of production type over time. Note that broiler farms exhibit earlier and higher incidence and the other production types are affected later. We consider this behaviour consistent with transmission spreading through poultry production networks.



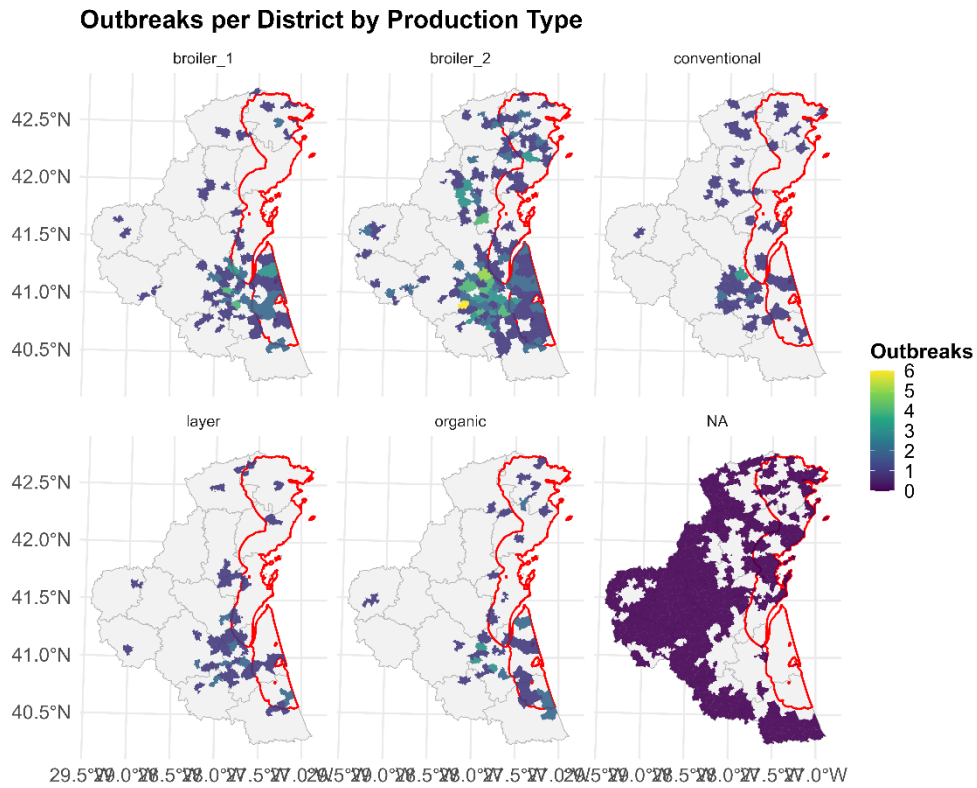


In the figure below we provide a spatial argument, aggregated over the entire period, to the disease spread. In panel (a) we show a spatial distribution of outbreaks by species: chickens and ducks. We see that there is a strong geographic clustering in the south-eastern coastal region, and a more scattered cluster in the northern region. We also note that the outbreaks of chickens dominate by number, but ducks also appear in the same spatial clusters. In panel (b) we have shown a spatial distribution by production type. We note that similar geographic cluster appears again across multiple production types: mostly driven by the broilers. In panel (c) we show outbreaks per district with incidence intensity. This confirms the clustering, and the outbreak is spatially heterogeneous. In panel (e) we show the outbreaks per county (which is important in our modelling approach) where the scale is coarser. The epidemic has an epicentre at Berks (yellow in colour). In panel (e) we show the county map with farm locations and density. We notice that outbreaks happen where the farm density is very high. The epidemic appears to be dependent on density of farms. Finally in panel (f) we portray farms within 1km of infected farms. This acts as an evidence to short-range transmission as many farms lie within 1 km of infected premises. We can say that the epidemic is behaving like a contact-network spatial epidemic.



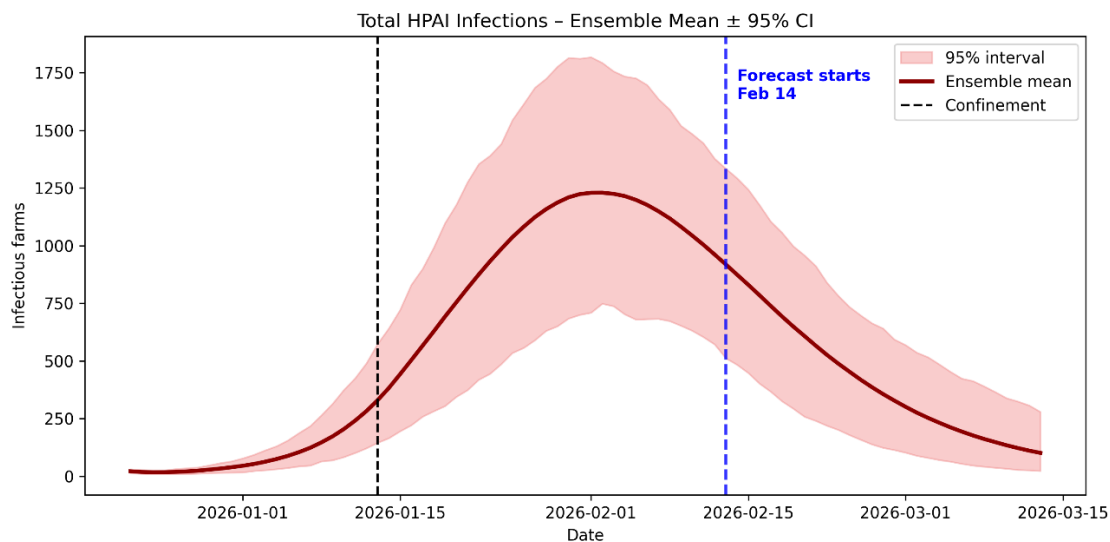


In the final plot, we show outbreaks per district per production type. This indicates where the disease is occurring and which production systems are driving transmission. We are aggregating outbreaks spatially in this plot. We see the strongest clustering in the broiler panels. Also, the outbreaks did not always lie inside or along the HRZ boundary, which is highlighted in red. There may be a need to re-define the HRZ.



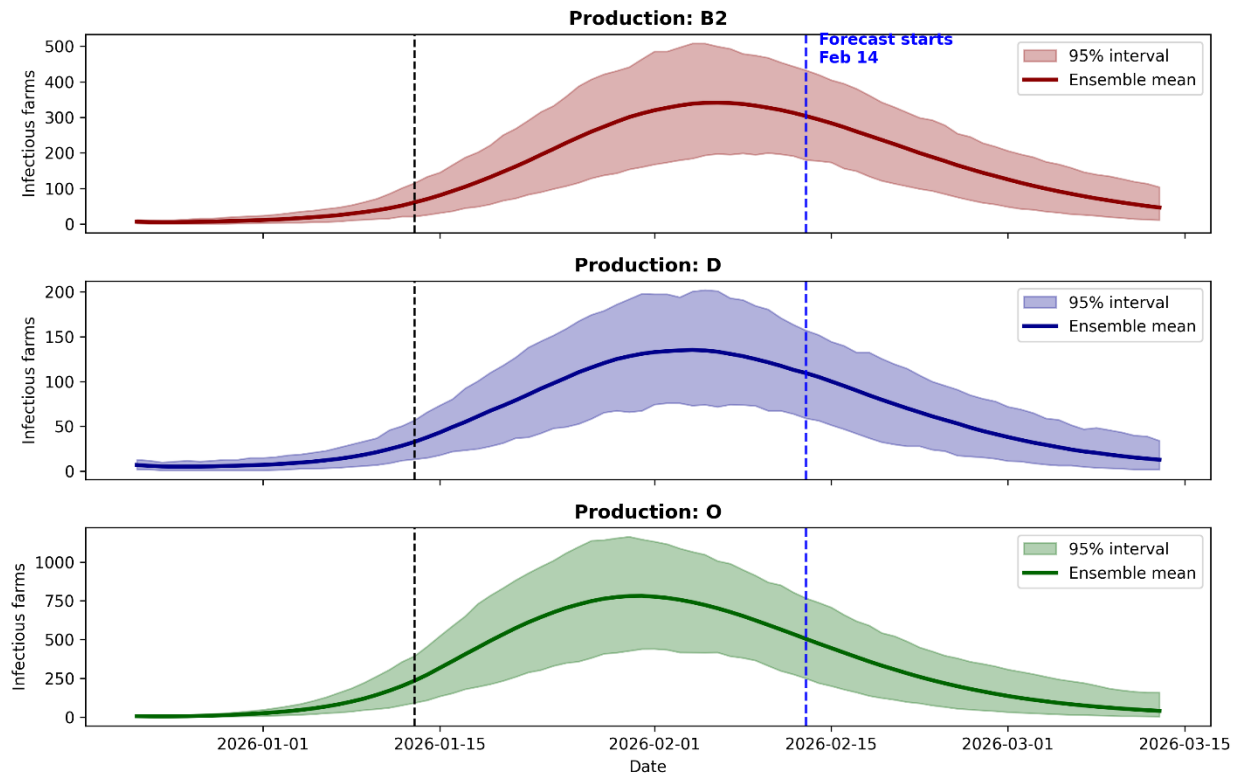
- ***A prediction of the likely temporal and spatial evolution of the epidemic over the following four weeks, assuming the management strategy remains as it is today; if you can have it by farm type, that would be nice.***

This analysis presents a forecast of the likely temporal and spatial evolution of the HPAI epidemic over the next four weeks, assuming the current management strategy remains unchanged. Projections are based on a stochastic, spatially explicit model that represents farm-level infection dynamics, reactive culling implemented at completion, preventive culling where applicable, and sector-specific confinement measures applied to broiler-2 (B2) and duck (D) production systems.



As seen in the total infection and production types timeseries, model results indicate that the epidemic has already passed its national peak, which occurred in early February, and is now in a declining phase. By the start of the forecast period in mid-February, total infections are decreasing, demonstrating that current control measures are sufficient to suppress further epidemic growth. However, the decline is gradual rather than rapid. Because farms continue to transmit infection until culling is completed, and environmental contamination persists beyond farm removal, new infections seeded earlier continue to contribute to transmission for several weeks. As a result, while the ensemble mean projects a steady reduction in infected farms, the model indicates that measurable infection is likely to persist throughout the four-week forecast horizon, with some residual infection still plausible into mid-March.

Ensemble Mean $\pm$ 95% Interval by Production Type

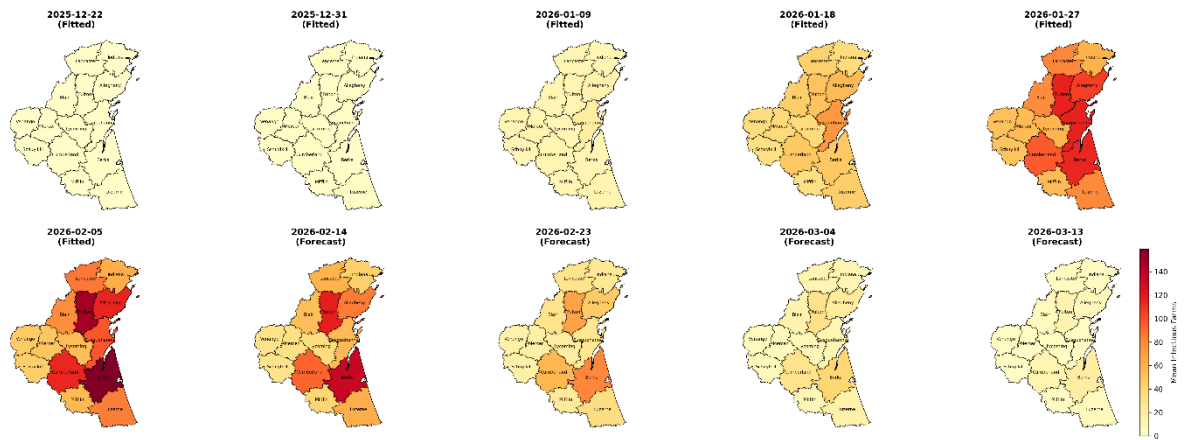


Spatial projections show a transition from widespread national transmission to increasingly localised infection. During the fitted period, infection spreads across many counties, forming multiple hotspots that align with the temporal peak. As the epidemic enters the forecast phase, the geographic footprint of infection contracts, and most counties experience substantial declines in infection intensity. Nevertheless, elimination is not uniform. A small number of counties remain persistently infected, sustaining the national epidemic tail. These residual hotspots are typically associated with regions containing higher densities of broiler and mixed production systems and reflect infections established before culling completion.

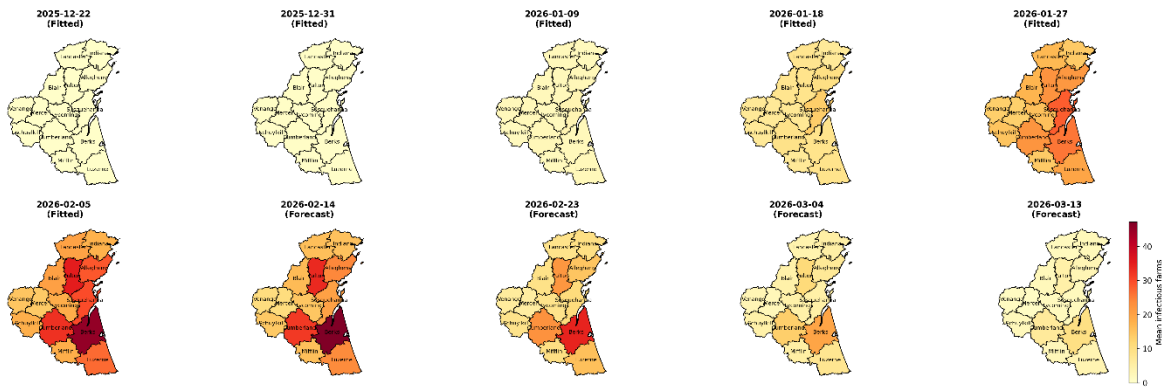
Based on the observation from the results, the epidemic is expected to continue declining over the next four weeks, but full elimination is unlikely within this period. Remaining risk becomes increasingly localised, with ongoing transmission concentrated in specific farm types and counties rather than across the entire country. Continued surveillance and response beyond the immediate forecast horizon will therefore be required to fully eliminate residual infection.

HPAI modelling challenge model description — second phase

Spatiotemporal HPAI Risk Progression (Ensemble Mean)
2025-12-22 to 2026-03-13

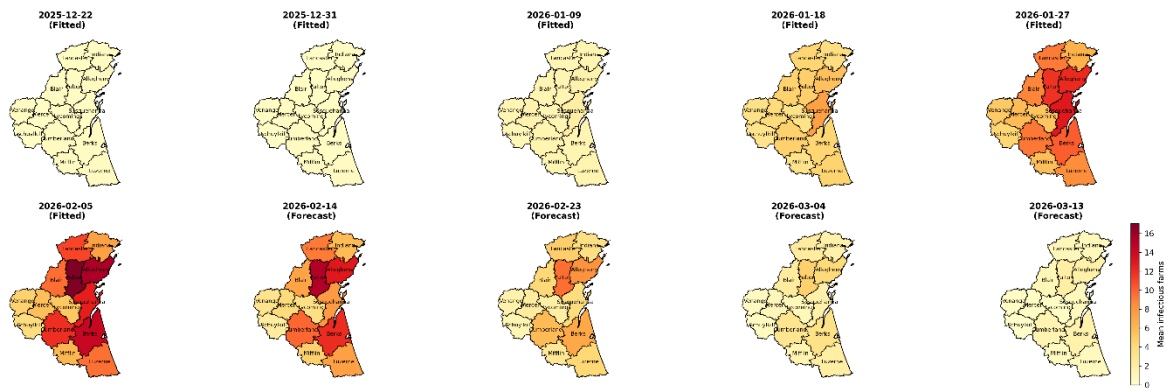


Spatiotemporal HPAI Risk Progression – B2
OPTION A (Culling at Completion Only)
Ensemble Mean · 10 Snapshots

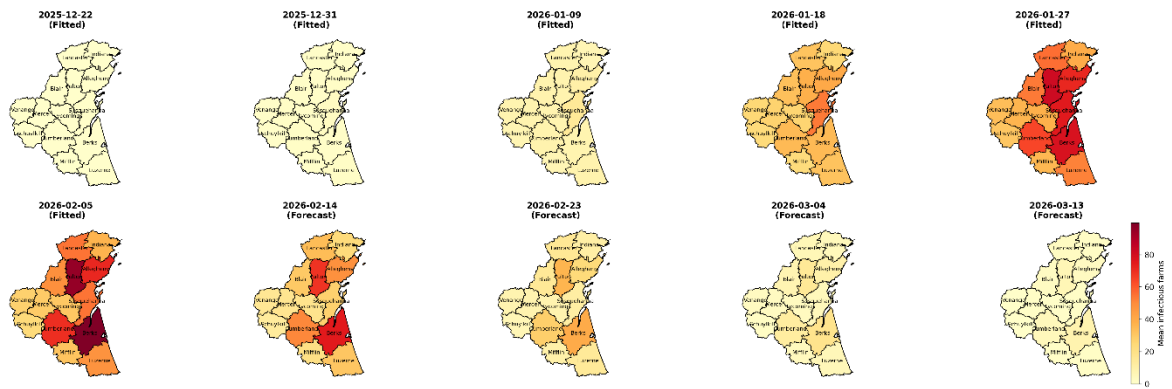


HPAI modelling challenge model description — second phase

Spatiotemporal HPAI Risk Progression – D
OPTION A (Culling at Completion Only)
Ensemble Mean · 10 Snapshots

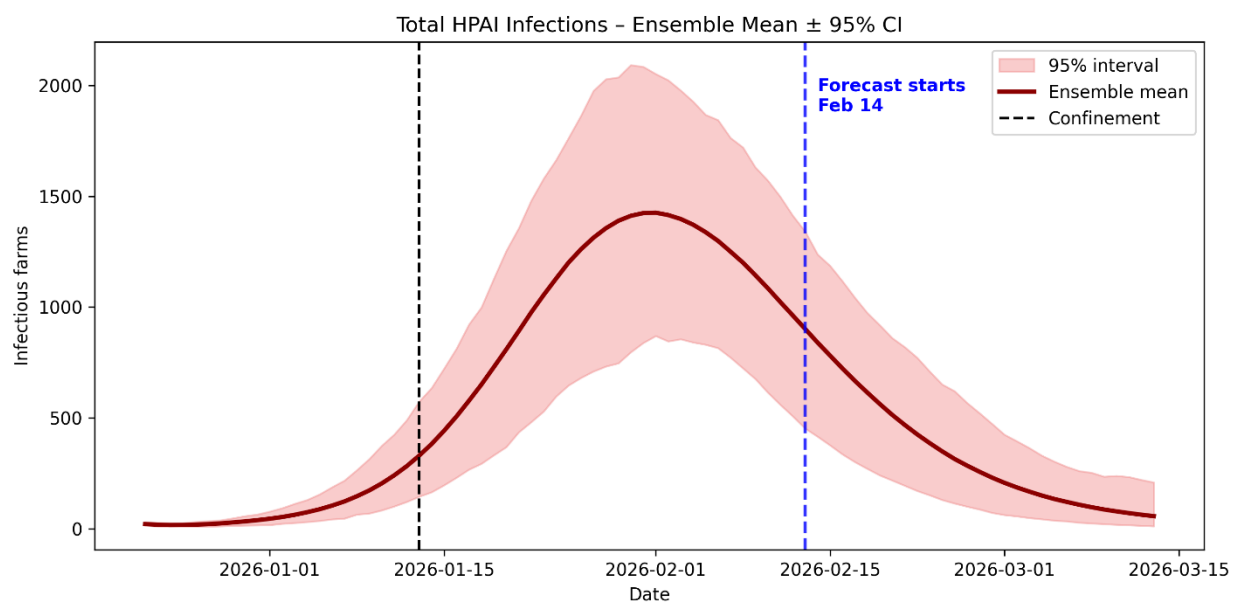


Spatiotemporal HPAI Risk Progression – O
OPTION A (Culling at Completion Only)
Ensemble Mean · 10 Snapshots



- ***An answer with a justification to the following question: given that duck farms likely to contribute less to the epidemic than chicken farms, what would be the epidemiological consequences of stopping the confinement of the organic ducks? And of stopping the confinement at all, even in chicken farms?***

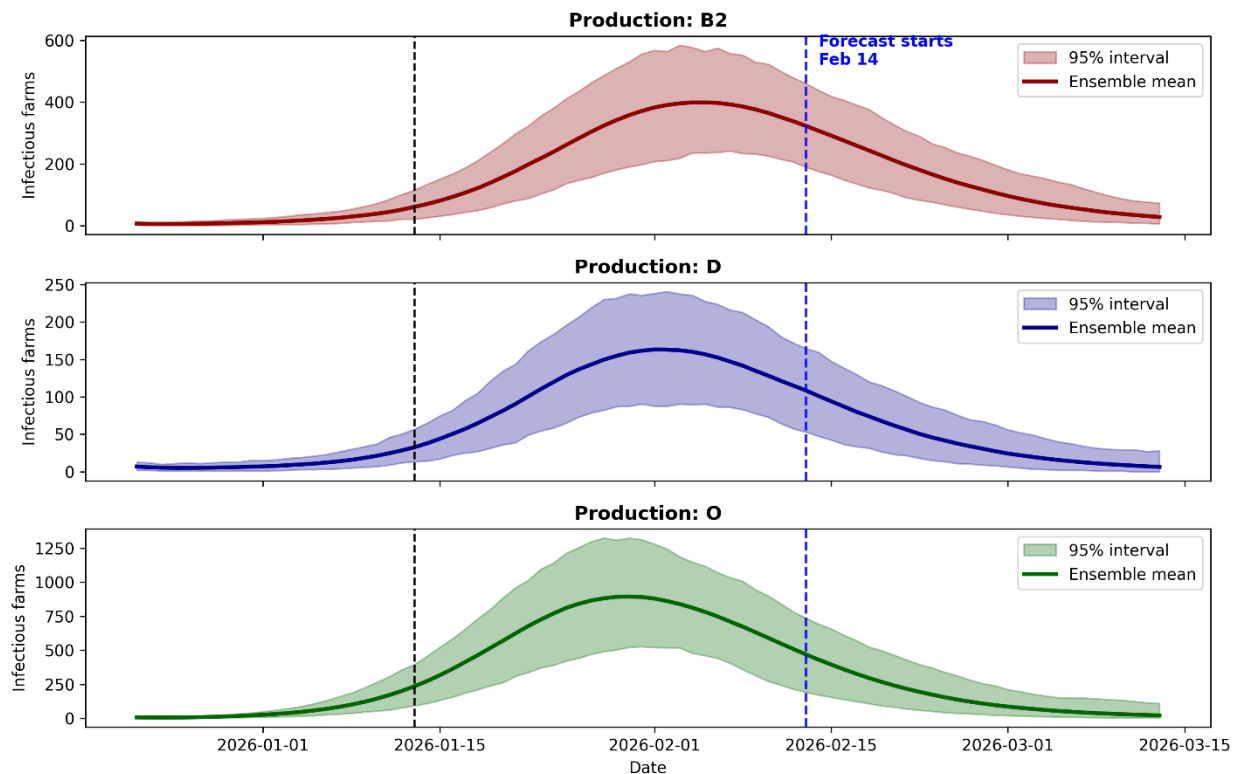
The plots below are results from when there is no confinement in the organic duck farms. Multiple confinement strategies were evaluated, including confinement applied to all production types, confinement applied selectively (e.g. chicken farms only or chicken and duck farms), and scenarios with no confinement at all. Across all scenarios, model outputs showed no material differences in key epidemic metrics, including peak incidence, timing of the peak, epidemic duration, and spatial extent of infection.



This finding indicates that, under the current model assumptions and parameterisation, confinement is not a binding control mechanism for epidemic dynamics. Instead, transmission is dominated by processes that confinement does not substantially affect.

First, the model demonstrates that early epidemic growth occurs before confinement can exert a meaningful effect. Reactive culling is applied only upon completion, with the earliest removals occurring around day three. A substantial proportion of transmission therefore occurs during the initial phase of the outbreak, when infected farms remain fully infectious and before confinement or culling completion can sufficiently reduce transmission.

Ensemble Mean $\pm$ 95% Interval by Production Type



Second, **within-county, within-production transmission—particularly in chicken production systems—is the dominant contributor to force of infection.** Confinement is implemented by reducing environmental exposure, environmental shedding, and movement-mediated transmission, but it does not alter direct within-production transmission. When this pathway drives epidemic growth, scaling down secondary transmission routes has limited impact on overall dynamics.

Third, **reactive culling overwhelms the marginal effects of confinement once it takes effect.** Reactive culling directly removes the most infectious units—outbreak farms—from the system, while confinement only moderates transmission from farms that remain infectious. As reactive culling is applied consistently across all scenarios, differences in confinement do not translate into measurable differences in outcomes.

Finally, duck farms contribute relatively little to epidemic amplification in the model. As a result, modifying duck-specific confinement has negligible influence on national-level dynamics, and extending confinement beyond chicken farms provides no additional epidemiological benefit. Overall, the similarity of outcomes across confinement scenarios reflects the dominance of early within-production transmission and delayed reactive culling.

Overall, the similarity of outcomes across confinement scenarios reflects the dominance of early within-production transmission and delayed reactive culling. Under these conditions, confinement measures act too late and on secondary pathways to substantially alter epidemic trajectories, highlighting the greater importance of rapid detection and faster completion of reactive culling for effective control.

Your opinion on your model

Strengths and limitations of your model

Strengths:

1. Our model is a mechanistic epidemiological model and not purely statistical.
2. We have implemented explicit control measures, and thus can be used to make decisions.
3. The network structure explains the multi-county spread and allows identification of regional amplification.
4. The model recognises chance events via multiple simulation runs, which are very relevant in farm epidemics.
5. The model is calibrated using the data provided meaning this can be used to advise policy makers having access to real world data.

Limitations

1. We have treated farms as identical units. However, in reality they can vary in size, security and other factors which are not taken into consideration
2. The model groups farms into counties rather than modelling individual farm locations. This leads to important transmission scenarios being neglected.
3. The model assumes outbreak detection and culling follow a fixed delay. This fixed delay can vary widely in real case scenarios.
4. Currently the comparison of preventive vs reactive culling strategies depends strongly on how preventive culling is done.

How much effort did it take you to complete this phase?

We divided the workload between two groups of three person each: one looking into mathematical modelling and the other looking into a statistical modelling approach. The sub-teams met at least four times during the whole phase. There were two meetings scheduled for the whole Flockbusters team to finalise, formalise, and decide on the model. Team leaders have spent around eight days working on the project during the phase. Each member of the rest of the contributing team have spent around four days on the project during the phase.

References

1. A modelling assessment of the impact of control measures on highly pathogenic avian influenza transmission in poultry in Great Britain Davis CN, Hill EM, Jewell CP, Rysava K, Thompson RN, et al. (2026) A modelling assessment of the impact of control measures on highly pathogenic avian influenza

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<https://doi.org/10.1371/journal.pcbi.1013874> 19

2. *Assessing the spatial and temporal risk of HPAIV transmission to Danish cattle via wild birds.* Chang, You, et al. (2025) *arXiv preprint arXiv:2504.12432*.

3. *A systematic review of mechanistic models used to study avian influenza virus transmission and control.* Lambert, S., Bauzile, B., Mugnier, A. et al. (2023) *Vet Res* **54**, 96. <https://doi.org/10.1186/s13567-023-01219-0>.

4. *Diseases, The Lancet Infectious.* "What is the pandemic potential of avian influenza A (H5N1)?" *The Lancet. Infectious diseases* 24.5 (2024): 437.